SPRINT 4 | DAYS 22-28

Streamlit Dashboard & Valuation Module Guide

Nifty 100 Financial Intelligence Platform

1. Module Overview

Sprint 4 delivers the interactive user-facing visualization and valuation analytics layers of the Nifty 100 Financial Intelligence Platform:

- **Module 5:** Streamlit Interactive Dashboard (8 screens)
- **Module 6:** Valuation Engine (`src/analytics/valuation.py`)

2. Streamlit Dashboard Architecture

The application follows Streamlit's multi-page application standard, under `src/dashboard/`:

- `app.py`: entry point configuring the wide layout and primary navigation.
- `utils/db.py`: centralized SQLite connector with `@st.cache_resource` connection caching and `@st.cache_data` query memoization.

Screen Breakdown

#	Screen / File	Features
01	**Home** `pages/01_home.py`	• High-level KPIs: Average ROE (%), Median P/E, Median D/E, Total Companies, Median 5Y Revenue CAGR, Debt-Free count • Dynamic sector breakdown distribution bar chart • Interactive year selector (2019-2024)
02	**Profile** `pages/02_profile.py`	• Company ticker selector and search • 10-year historical Revenue and Net Profit bar charts • Profitability trend lines (ROE %, ROCE %, OPM %) • Balance sheet breakdown (Equity, Reserves, Borrowings, Other Liabilities)
03	**Screener** `pages/03_screener.py`	• Custom metric sliders: ROE, D/E, FCF, Revenue CAGR, PAT CAGR, OPM, P/E, P/B, Dividend, ICR • 6 instant presets: Quality, Value, Growth, Dividend, Debt-Free, Turnaround • Live ranked results table and CSV export
04	**Peers** `pages/04_peers.py`	• Peer group benchmarking across 11 designated industry cohorts • Radar charts comparing company percentiles against peer group medians

#	Screen / File	Features
05	**Trends** `pages/05_trends.py`	• Multi-metric historical comparisons and macro sector trend overlays
06	**Sectors** `pages/06_sectors.py`	• Sector-wide median KPIs, index weight analysis, and valuation distribution
07	**Capital** `pages/07_capital.py`	• Visual capital allocation map classifying companies across the 8 cash flow patterns
08	**Reports** `pages/08_reports.py`	• Annual report repository and document links per company

3. Valuation Engine

The engine (`src/analytics/valuation.py`) computes multi-year valuation multiples and flags anomalies.

- **Multiples:** P/E, P/B, EV/EBITDA.
- **FCF Yield:** `(Free Cash Flow Cr / Market Cap Cr) * 100`
- **Benchmark comparables:** 5-Year Median P/E and Sector Median P/E.

Valuation Flags

Flag	Rule	Meaning
Caution	Company P/E > 1.5x Sector Median P/E	Overvalued
Discount	Company P/E < 0.7x Sector Median P/E	Undervalued / value opportunity
Fair	Trading within 0.7x to 1.5x of sector median	In line with sector

Outputs

Output	Details
`output/valuation_summary.xlsx`	Complete 92-company valuation table
`output/valuation_flags.csv`	44 flagged companies (30 Discount, 14 Caution)
SQLite table	`valuation` table in `nifty100.db`

4. Launch Instructions

```
cd C:\Users\USER\Downloads\intern\w3
streamlit run src/dashboard/app.py
```

The app runs locally at `http://localhost:8501`.